

Positive reviews (n = 15439):

9.29% 1435 Bad service, wrong orders, rude staff
32.63% 5038 Fast and quick service, clean place, friendly staff
35.78% 5524 Overall good price, quality, service
7.51% 1160 Food quality issues (cold/old fries, burgers, nuggets)
8.02% 1238 Long wait times, slow drive-thru
6.76% 1044 Miscellaneous, general complaints about environment

Negative reviews (n = 12139):

40.01% 4857 Bad service, wrong orders, rude staff
7.71% 936 Fast and quick service, clean place, friendly staff
2.75% 334 Overall good price, quality, service
14.78% 1794 Food quality issues (cold/old fries, burgers, nuggets)
25.55% 3102 Long wait times, slow drive-thru
9.19% 1116 Miscellaneous, general complaints about environment

Chi-squared (all 6 topics vs uniform):

Positive: $\chi^2 = 8626.066$, $p = 0.0000 \rightarrow$ Reject uniform

Negative: $\chi^2 = 6971.831$, $p = 0.0000 \rightarrow$ Reject uniform

One-Proportion Z-Tests:

Hypotheses: $H_0: p \geq 0.30$ vs $H_a: p < 0.30$

Significance level: $\alpha = 0.05$

(Left-tailed: we reject H_0 only if the proportion is significantly below 30%.)

Test 1: Positive reviews (quality of food topic)

Topic used: "Overall good price, quality, service"

Count in topic: $x = 5524$ out of $n = 15439$ positive reviews

Sample proportion: $\hat{p} = 0.3578$ (35.78%)

z-statistic: 15.6708 p-value: 1.0000

Conclusion: Do not reject H_0

Test 2: Negative reviews (customer service topic)

Topic used: "Bad service, wrong orders, rude staff"

Count in topic: $x = 4857$ out of $n = 12139$ negative reviews

Sample proportion: $\hat{p} = 0.4001$ (40.01%)

z-statistic: 24.0703 p-value: 1.0000

Conclusion: Do not reject H_0

We do not have evidence that the proportion is below 30% (at least as expected).

Positive reviews (food quality): proportion 35.78% is not significantly below 30% ($\alpha=0.05$).

Negative reviews (customer svc): proportion 40.01% is not significantly below 30% ($\alpha=0.05$).

So the areas of interest (food quality in positive reviews, customer service in negative)

both perform at least as expected: we cannot say either proportion is below 30%.